V.L Memorial Public School

475 Vijay Nagar

CHAPTER-5-QUESTIONS-CLASS7TH-SCIENCE

Class 07 - Science (Curiosity)

Section A

1. Which of the following is a physical change? **[1]**
   - Burning of candle
   - Cooking of egg
   - Melting of chocolate
   - Rusting of iron

   a) Cooking of egg
   b) Rusting of iron
   c) Melting of chocolate
   d) Burning of candle

2. Which of the following is a chemical change? **[1]**
   - Melting of ice
   - Evaporation of water
   - Souring of milk
   - Dissolving sugar in water

   a) Souring of milk
   b) Dissolving sugar in water
   c) Evaporation of water
   d) Melting of ice

3. What type of change occurs when only the shape, size, or state of a substance changes? **[1]**

   a) Nuclear change
   b) Chemical change
   c) Biological change
   d) Physical change

4. Which of the following is a physical change ? **[1]**

   a) Melting of wax
   b) Combustion of magnesium ribbon
   c) Burning of candle
   d) Rusting of iron

5. Which type of change occurs when erosion is caused by flowing water or wind? **[1]**

   a) Biological change
   b) Nuclear change
   c) Chemical change
   d) Physical change

6. Which of the following is a chemical change? **[1]**
   - Melting of wax
   - Freezing of water
   - Boiling of water
   - Digestion of food

a) Digestion of food
b) Boiling of water
c) Melting of wax
d) Freezing of water

7. Which of the following is a physical change? **[1]**

- Burning of wood
- Cooking of rice
- Tearing of paper
- Rusting of iron

a) Tearing of paper
b) Burning of wood
c) Cooking of rice
d) Rusting of iron

8. Which of the following statements is true for a physical change? **[1]**

a) It always produces heat
b) A new substance is always formed
c) It is always irreversible
d) It involves change in physical properties

9. What is the process called when wind or flowing water moves rock materials from one place to another? **[1]**

a) Weathering
b) Combustion
c) Erosion
d) Rusting

10. Which of the following is a chemical change? **[1]**

- Cutting of wood
- Melting of wax
- Burning of wood
- Boiling of water

a) Melting of wax
b) Cutting of wood
c) Burning of wood
d) Boiling of water

11. In which type of change a new substance is formed? **[1]**

a) Change in size only
b) Both Physical change and Change in size only
c) Physical change
d) Chemical change

12. Which process produces light without producing heat in living organisms? **[1]**

a) Bioluminescence process
b) Rusting process
c) Oxidation process
d) Combustion process

13. What are the heaps of sand, soil, and stones formed by the breaking of rocks called? **[1]**

a) Minerals
b) Crystals
c) Sediments
d) Fossils

14. Which substance is described as combustible? **[1]**

a) Water stored indoors
b) Wood used commonly

c) Glass kept safely
d) Sand placed nearby

15. What is a characteristic of a physical change? **[1]**

a) Irreversibility
b) Formation of a new substance
c) Release of gas
d) Change in physical properties without forming a new substance

16. Which of the following is not a characteristic of a physical change? **[1]**

a) Formation of a new substance
b) Change is easily reversible
c) Temporary change
d) No new substance is formed

17. What is a chemical change? **[1]**

a) A change in colour
b) A change in position
c) Formation of new substances
d) A change in shape

18. Which gas is produced when zinc reacts with dilute sulphuric acid? **[1]**

a) Hydrogen
b) Carbon dioxide
c) Oxygen
d) Nitrogen

19. Which of the following is a sign of a chemical change? **[1]**

a) Formation of a precipitate
b) Change in shape
c) Change in state
d) Change in size

20. What are changes that can return to the original form called? **[1]**

a) Permanent changes
b) Irreversible changes
c) Unrelated changes
d) Reversible changes

21. **Assertion (A)**: Rusting of iron is a physical change. **[1]**
Reason (R): It only changes the appearance of iron.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Carbon dioxide gas supports combustion. **[1]**
Reason (R): Wood, cotton, paper and kerosene are the examples of combustible substances.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Tearing of paper is a chemical change. **[1]**
Reason (R): It changes the size and shape of the paper.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** Change of water from liquid to steam on heating is a physical change. **[1]**
Reason (R): The state of water during heating remains the same and the change involves a change in composition.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

25. **Assertion (A):** Boiling of water is a chemical change. **[1]**
Reason (R): It changes water into steam.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

26. **Assertion (A):** Souring of milk is a chemical change. **[1]**
Reason (R): It results in the formation of new substances.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

27. **Assertion (A):** Freezing of water is a physical change. **[1]**
Reason (R): It involves a change in state without forming a new substance.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

28. Define a physical change. **[1]**
29. Is dissolving sugar in water a physical or chemical change? **[1]**
30. What type of change is called in which new substance with new properties is formed? **[1]**
31. Why is the rusting of iron considered a chemical change? **[1]**
32. Explain what is meant by a desirable change, giving one example from daily life. **[1]**
33. What are the two groups into which changes around us can be classified? **[1]**
34. Gives examples of natural slow changes. **[1]**

Section B

35. Match the following: **[2]**

Change	Reversible or Irreversible
(a) Boiling of milk	(i) Reversible
(b) Breaking of a glass	(ii) Irreversible
(c) Freezing of water	(iii) Reversible
(d) Souring of curd	(iv) Irreversible

36. Match the following: **[2]**

Column A	Column B

(a) Ripening of fruits	(i) Quick and desirable physical change
(b) Decay of vegetables	(ii) Slow, natural, undesirable chemical change
(c) Cutting of vegetables	(iii) Slow, natural and desirable change
(d) Rusting of iron	(iv) Slow, natural but undesirable change

37. Are the changes of water to ice and water to steam, physical or chemical? Explain. **[2]**
38. Explain why erosion is considered a physical change. **[2]**
39. Is melting of wax a physical or chemical change? Explain. **[2]**
40. What happens when magnesium ribbon is burnt? **[2]**
41. How can rusting be prevented? **[2]**
42. Explain why the candle covered with a glass tumbler stops burning. **[2]**
43. Is cooking of food a chemical change? Why? **[2]**

Section C

44. Differentiate rusting from the burning of magnesium ribbon as chemical changes. **[3]**
45. Explain the changes observed when a candle burns. **[3]**
46. Describe an activity to show that burning of paper is a chemical change. **[3]**
47. Explain why stretching a rubber band is a physical change. **[3]**
48. How does continuous flowing water change the appearance of river rocks and pebbles? **[3]**
49. Explain how focused sunrays cause paper to catch fire. **[3]**
50. How can lime water be used to test for the presence of carbon dioxide in a gas? **[3]**

Section D

51. **Read the following text carefully and answer the questions that follow:** **[4]**

Many useful changes happen in our daily life for example, the changing of milk into curd, ripening of fruits, cutting of fruits, and cooking of food. All these are desirable changes. Many useful changes happen in our daily life. For example, the changing of milk into curd, ripening of fruits, cutting of fruits, and cooking of food. All these are desirable changes. On the other hand, some changes may be undesirable, such as the rusting of iron or the decay of food during its storage. A change that is undesirable in some situations may be desirable in other situations. For example, decomposition of food can be very useful in converting food waste into compost. Some changes occurring over the years due to human activity can have a long-term environmental impact. For example, the increased consumption of fuels in cars, trains, aeroplanes, etc., is increasing the amount of carbon dioxide in the atmosphere. Drying of paint on walls, doors, furniture, etc., releases many substances through evaporation, causing atmospheric pollution.

Questions:

a. Give an example of a desirable change. (1)
b. Why is the rusting of iron considered an undesirable change? (1)
c. How can the decomposition of food be both undesirable and desirable? (2)

OR

How does the increased consumption of fuels affect the environment? (2)

52. **Read the following text carefully and answer the questions that follow:** **[4]**

In the rusting of iron a new brown-coloured substance called rust is formed. Thus, rusting is also a chemical change because it involves the formation of a new substance, iron oxide. A chemical reaction in which a substance reacts with oxygen and produces heat and/or light is called combustion. Substances that undergo combustion reactions are called combustible substances. For example, wood, paper, cotton, kerosene, etc., are combustible substances. Since the candle covered by the glass tumbler does not have a continuous supply of air, the flame gets extinguished soon. The component of air that supports combustion is oxygen. This can be confirmed by the presence of carbon dioxide in the glass tumbler inverted on the candle.

Questions:

a. Why is rusting of iron considered a chemical change? (1)

b. What component of air supports combustion? (1)

c. What happens when a candle is covered with a glass tumbler? Explain why the flame gets extinguished. (2)

OR

What is combustion? Name any two combustible substances. (2)

53. **Read the following text carefully and answer the questions that follow:** **[4]**

We have noticed fine sand collecting on riverbeds or in lakes, the sand formed when rock pebbles, soil, and sediments are broken down and moved from one location to another by natural forces like wind and flowing water. This process is called erosion. Erosion during a landslide is an example of a physical change. River rocks and pebbles often appear smoother due to the constant erosion caused by the flowing water. Erosion caused by flowing water and wind is a physical change. When the speed of the water or wind decreases, such as in an ocean or a lake, the material transported during erosion settles down at the bottom. These sediments harden over time and become new rocks. Most of these changes take place over thousands of years and cannot be reversed.

Questions:

i. What is erosion? (1)

ii. How do river rocks and pebbles appear after erosion? (1)

iii. Explain how sediments are formed during the process of erosion. (2)

OR

Why is erosion caused by flowing water considered a physical change? (2)

Section E

54. Evaluate how weathering contributes to soil formation through physical and chemical changes. **[5]**

55. Describe an experiment to demonstrate that burning a candle involves both physical and chemical changes. **[5]**

56. Compare and contrast the changes involved in melting ice and burning paper. **[5]**

57. How can you prove that dissolving salt in water is a physical change? **[5]**

58. Compare weathering and erosion, including their processes and outcomes. **[5]**

Section F

59. **State whether the given statement is True or False:** **[8]**

(a) Making a solution of lemon juice and water is a physical change. **[1]**

(b) Burning of paper is a chemical change. **[1]**

(c) Mixing baking soda with lemon juice is a chemical change. **[1]**

(d) Physical changes are usually reversible. **[1]**

(e) A change is called reversible if we can get back the original substance. **[1]**

(f) Dissolving salt in water is a chemical change. **[1]**

(g) Formation of clouds is a reversible change. **[1]**

(h) Burning of candle involves physical change only. **[1]**

Section G

60. **Fill in the blanks:** **[9]**

(a) Cooking of food is a _____ change. **[1]**

(b) A change in which no new substance is formed is called a _____ change. **[1]**

(c) The mixing of baking soda with lemon juice produces bubbles due to the release of _____ gas. **[1]**

(d) Melting of ice is a _____ change. **[1]**

(e) Magnesium, when burnt in air, produces a substance called _____. The substance formed is _____ in nature. Burning of magnesium is a _____ change. **[1]**

(f) A change in which a new substance is formed is known as a _____ change. **[1]**

(g) Boiling of water is a _____ change. **[1]**

(h) To show that a chemical change has occurred, we can look for signs like gas formation, colour change, and Change in _____. **[1]**

(i) Nalini observed that the handle of her cycle has got brown deposits. The brown deposits are due to _____ and this is a _____ change. **[1]**